CS 598 Research Project Report: Group-7

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Abstract

More and more developers are writing software solely through vibe coding, building entire programs by chatting with a coding agent. Over a long session, the developer will place many constraints upon the code that the agent then agrees upon; each new request can break one that they agreed to earlier. Vibe coders rarely read the generated code, so these broken constraints may go unnoticed. How can we gain confidence that vibe-coded software actually does what the developer asked for? Historically, formal verification has offered such guarantees that regular software testing cannot, and recently LLMs have made formal verification much more accessible. However, current approaches either audit code offline or merely fix a specification before any code is written—none of them track constraints during a session. We propose a Lean companion for vibe coding that does exactly this. Our companion listens to the developer-agent conversation, formalizes the agreed-upon constraints as Lean invariants, and builds a model of the code in Lean. After every turn of the coding agent, the companion checks every short sequence of actions in the model against these invariants and returns failed checks to the agent, thus catching broken constraints before the developer has to.

1 Artifact Information

Link to the GitHub Repository: https://github.com/Rohpar1504/CS-598-Research-Project-Group-7

2 Introduction

Large language models are being rapidly deployed as fully autonomous agents. As these agents' capabilities have grown over the past few years, more developers are using this autonomy to rapidly prototype code via vibe coding—writing software completely by chatting with an agent. While vibe coding lets developers generate more code than ever before, the iterative back-and-forth of the conversation places many constraints on the generated code. In particular, these constraints might disagree with each other as they evolve over time, resulting in significant technical debt for the agent, and more importantly, the developer. Worse yet, LLM-generated code has been known to contain security flaws [4], and since the developer is vibe coding, they are less likely to notice flaws. This raises the question: how can we gain confidence over vibe-coded software?

Formal verification is one such way we can acquire guarantees on the properties of a program, and it offers a unique certainty of correctness that simple testing cannot. Tools such as proof assistants (e.g. Lean [3], Rocq [16]) and specification languages (e.g. TLA+ [10]) are often used for formal verification, but have historically required too much expertise and development time to justify its use. LLMs have significantly lowered both costs. For example, Specula [1] uses agents to write TLA+ models of large systems and check them for bugs, and Quint provides tooling for agents to write executable specifications from requirements or source code [14].

None of the existing techniques track constraints during a vibe-coding session. Current approaches either are purely offline audits, or specification-driven tools that merely fix requirements before any of the code generation begins. Mitchell and Shaaban [11] propose a formal side-car that would track these constraints, but their prototype only performs lightweight syntactic checks.

We propose a Lean companion that runs in the background during a vibe-coding session. The companion will listen to the developer-agent conversation and formalize the constraints the developer and agent agree upon as Lean invariants, as well as maintain a Lean model of the generated code for checking. This process repeats after every turn of the coding agent.

We evaluate the companion on the stateful problems of SlopCodeBench [12] and EvoCode-Bench [15], augmented with a simulated developer. We compare its bounded model checking against property-based tests generated from the same invariants, both on seeded mutants of the reference solutions and on regressions from our agent runs. End-to-end sessions measure how accurately it extracts invariants and how much overhead it adds, as well as testing whether its feedback reduces regressions relative to four baselines.

3 Proposed Technique

3.1 Approach Overview

Our design is a companion for vibe coding. The companion runs alongside a coding agent during a session while maintaining a model of the application's state in Lean. It checks this model against invariants formalized from the conversation and runs after every turn of the coding agent. The companion operates in four stages (Figure 1):

(1) *Constraint Extraction.* The companion translates constraints stated by the developer into Lean invariants. One example of such a constraint is "There should be no double-booking of a room". These invariants are confirmed by the developer through examples.
(2) *Lean Model.* An LLM updates a Lean model of how the code's actions change the application's state. Trace validation runs the same sequence of actions on the model and the code via a test harness that can map each action the model takes to a call into the code. The LLM then repairs the model if the resulting states differ.
(3) *Invariant Checking.* The companion performs bounded model checking, exhaustively checking every action sequence up to a fixed length in the model against the invariants.
(4) *Feedback.* If model checking finds a counterexample, a sequence that violates an invariant, then we translate that

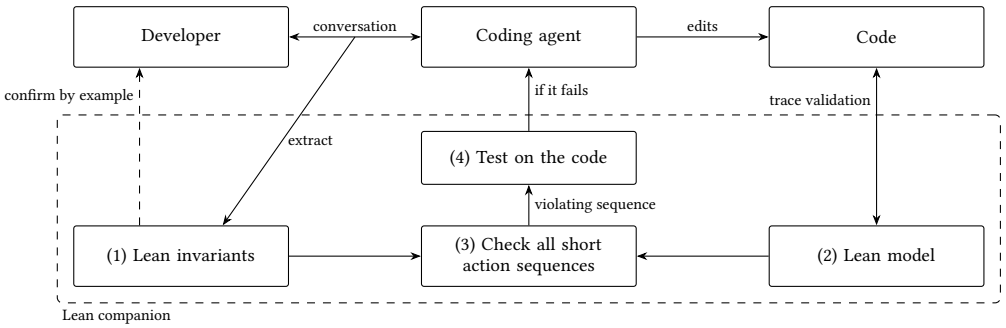


Figure 1: Overview of the Lean companion.

counterexample into a test on the code. If that test fails, then the result is returned to the coding agent.

4 Experimental Evaluation

No existing benchmark combines multi-turn evolution, persistent application state, constraints that carry across turns, and publicly runnable tests. Therefore, we adapt two existing benchmarks: SlopCodeBench [12] and EvoCode-Bench [15]. We use a simulated developer that supplies each change to the specification conversationally, and we use the checkpoint specifications as the ground truth. Agent runs are also noisy and expensive, so we split the evaluation into a controlled detection study that runs no agents and a study of end-to-end sessions. RQ2 draws on both studies, while RQ1, RQ3, and RQ4 are only answered by the end-to-end sessions. We organize the evaluation around four research questions:

RQ1 (Formalization fidelity) How accurately does the companion translate constraints stated in the developer-agent conversation into Lean invariants, and how reliably does it update or retire constraints that are later revised or contradicted?

RQ2 (Violation detection) Given the same invariants and the same time budget, does bounded model checking over the Lean model find violations that property-based tests on the code miss?

RQ3 (Regression mitigation) When the detected violations are returned to the coding agent, does the companion reduce regressions over a multi-turn session, compared with an unassisted agent, an agent reminded of the constraints, a PBT side-car, and an oracle gate?

RQ4 (Overhead) What latency and token overhead, and what failure rate, do model synthesis, model repair, and checking produce per turn, and how do these scale as the session and the generated code grow?

4.1 Evaluation Setting

Benchmarks. Our primary benchmark is the stateful subset of SlopCodeBench [12], where the Python problems evolve over a series of checkpoints and then re-run the tests from all earlier checkpoints at each one as a regression oracle. Unlike function-level suites [6], these problems maintain persistent state governed by rules. Additionally, as a secondary benchmark, we use a subset of EvoCode-Bench [15], for which the longer tasks include corrections and conflicts that supersede earlier requirements.

Simulated developer. An LLM-based simulated developer delivers each checkpoint's specification change conversationally, paraphrasing it, splitting it across messages, or leaving it implicit. It also answers the companion's confirmation questions according to the specification. The checkpoint specifications are fixed across configurations and serve as ground truth.

Controlled detection. To test the checking approaches without agent-run variance, we inject faults into the reference solution for the benchmark problems by using mutation operators modeled on common agent regressions: dropping a guard, performing an update before the validation check, an off-by-one error on a boundary, and a change in the output format. Every such mutant breaks one of the known constraints of the problem specification. On these mutants and on the regressions produced by our own agent runs, we compare property-based testing of the program code to the bounded model checking performed by the Lean companion, using the same invariants and the same time budget. The unmutated reference solutions measure false positives.

Configurations. All configurations use Claude Code as the coding agent and share a one-retry budget per checkpoint: (1) agent-only, with no side-car; (2) a constraint reminder, which places the extracted constraints in the agent's context at every turn without checking them; (3) a PBT side-car, which translates our extracted invariants into property-based tests [7] and runs them on the code through the same test harness as the companion, so that only the checking method differs; (4) our Lean companion, which checks the invariants against a Lean model of the code, written and repaired by an LLM and validated against the code by trace validation, and returns failing tests to the agent; and (5) an oracle gate, modeled on the Verification Gate of Huang et al. [6], which reruns the oracle and rolls back on regression, and which serves as an upper bound since it has access to ground truth. The PBT side-car and the Lean companion receive the same time budget per check. Each configuration is run three times on each task.

Metrics. For RQ1, we measure precision and recall of generated invariants against the constraints in the checkpoint specifications, with and without developer confirmation, and the fraction that typecheck in Lean. If a constraint is revised or contradicted by a

checkpoint, we also measure how many of the superseded constraints the companion correctly updated or retired, as well as the number of stale invariants that are still active. For RQ2, we measure detection recall and precision against the seeded mutants and the agent-run regressions that violate a stated constraint, and turns-to-detection for agent-run regressions. For RQ3, following Huang et al. [6], we report the per-checkpoint oracle pass rate, the degradation from the first to the final checkpoint, and the fraction of problems that regress. We also report the rate at which requirements that once passed are broken later, since a single persistent failure makes a whole checkpoint fail. For RQ4, we measure the time and cost of writing and repairing the model, the added latency and tokens per turn, and the rate of failed or timed-out checks.

5 Related Work

Our work builds on three areas of research: formal side-cars for vibe coding, model extraction and conformance checking, and intent formalization.

Formal Side-cars. Mitchell and Shaaban [11] argue for "Vibe Reasoning" to go along with vibe coding in their paper; in particular, they argue that vibe coding can suffer from a breakdown in the reconciling of user constraints over a long coding session. To this end, they propose a side-car that automatically formalizes specifications from the conversation and validates the code against them. If there is a failure in the check, then this is returned to the coding agent. Our system is an implementation of this side-car with a semantic model and exhaustive checking. Similarly, VibeContract [17] breaks down the intent of the natural language into specific tasks with generated contracts. These contracts then guide the validation and debugging of the program. Forge [18] is a closed-loop pipeline that extracts models from source code and checks them with a verifier. However, this pipeline only works with a restricted subset of Java so that they can extract their models.

Model Extraction and Conformance. Specula [1] is a tool that generates formal specifications in TLA+ for large code repositories, and validates the specifications against execution traces. However, it only runs offline, takes hours per repository, and importantly generates invariants mostly from implemented code, so that a constraint the code already violates may never be translated into an invariant. We likewise use trace validation. Quint [13] is an executable specification language based on TLA that is equipped with a model checker. Importantly, its tooling allows agents to generate Quint specifications from requirements or given source code [14]. Developers can invoke these tools on demand, while our companion maintains its specification model automatically during a vibe coding session. One tool built on Quint is FuzzMo [2]. FuzzMo uses an LLM to synthesize Quint models from code and then checks conformance by comparing Quint execution traces against the implemented code. The authors distinguish code-based models, which are specifications that match the implementation, from ideal-based models, which are specifications that match what the software should do. Our design is a hybrid of these two approaches; our specification model follows the code while the invariants that the model is checked against come from the conversation.

Intent Formalization. Much of our approach relies on recognizing the intent of a user throughout a vibe coding session. Lahiri [8] argues that translating this intent into formal specifications is an important challenge for LLM-generated code, and that only the user can judge the correctness of such specifications. We address one instance of this by extracting invariants from the vibe coding conversation. TiCoder [9] uses user feedback on generated tests to formalize intent. Similarly, we confirm extracted invariants with the developer. nl2postcond [5] studies using LLMs to translate code documentation into postconditions. The authors evaluate the postconditions' correctness and how well they distinguish incorrect code. However, this only targets individual functions, not an entire program's state. In industry there are specification-driven tools such as Kiro [7], which prompt an agent to write a specification before writing any code. Kiro translates the properties from its specification into property-based tests. Still, this tool does not check these specifications against a formal model.

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